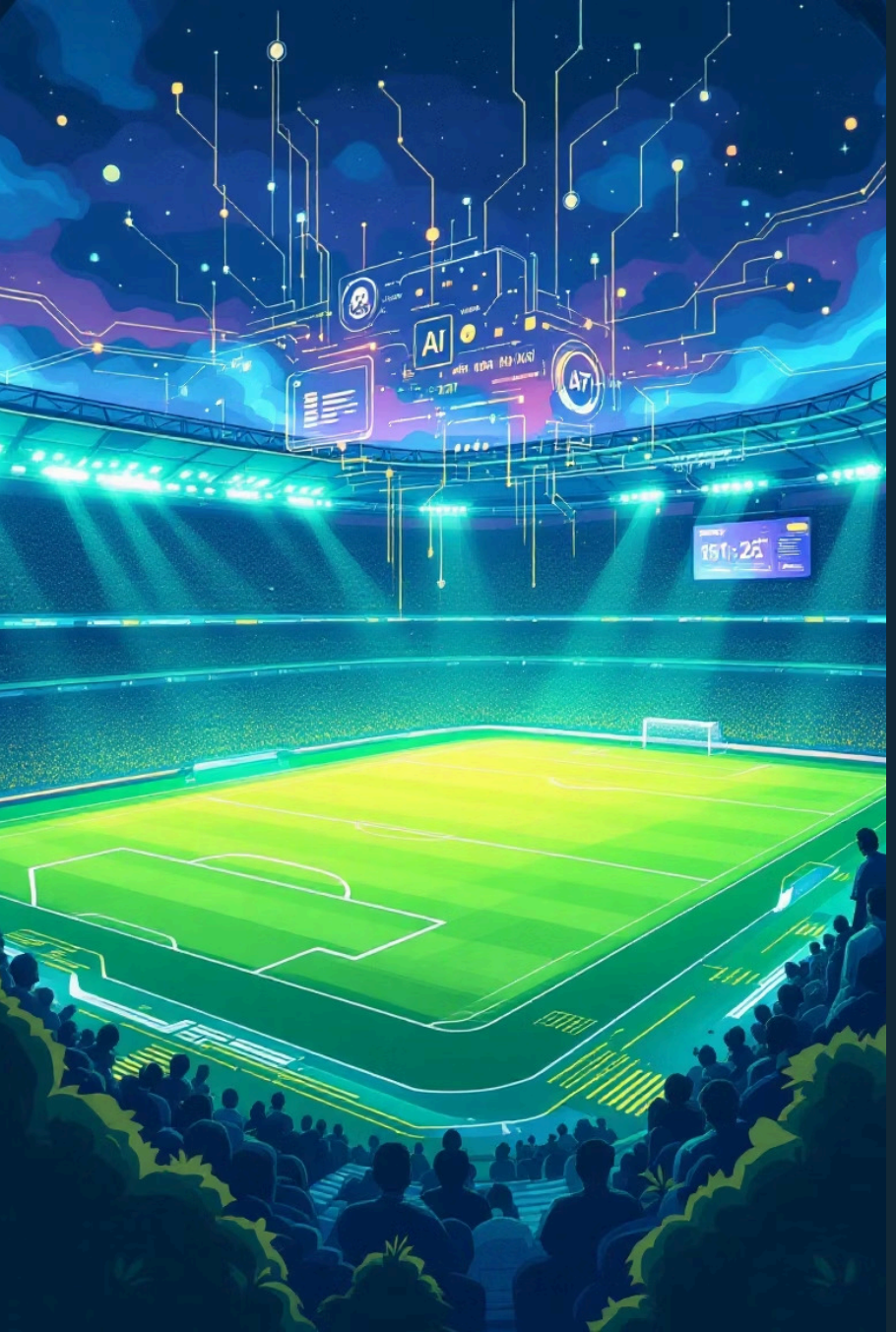


TransferIQ: AI-Driven Player Transfer Value Prediction

Dynamic Forecasting using AI and Multi-source Data

PRESENTED BY: ENUGULA AKSHITHA

Project Overview: Predicting Player Transfer Values

Estimating a player's transfer value is a complex task influenced by performance, age, team, public sentiment, and market trends. This project proposes an AI-driven model to predict these values by integrating multi-source data.

AI-Driven Model

Leveraging advanced machine learning and time-series forecasting.

Multi-Source Data

Integrating performance stats, social media sentiment, injury history, and market data.

Dynamic Predictions

Providing a data-driven approach to forecasting transfer values.

Methodology: A Staged Approach

The project is structured into several stages, applying diverse AI techniques to predict transfer values, with a focus on multi-source data integration and time-series forecasting.

01

Data Collection & Preprocessing

Gathering and cleaning diverse datasets.

02

Model Development

Building LSTM and ensemble models.

03

Time-Series Forecasting

Utilizing LSTMs for temporal dependencies.

04

Sentiment Analysis

Quantifying public perception's impact.

Data Collection & Preprocessing

A critical first step involves gathering and preparing data from various sources to ensure comprehensive and accurate model inputs.

Data Sources

- Player Performance data from GitHub
- Transfermarkt (Market Value)
- Social Media Sentiment data from GitHub
- Historical Player Injury Records

Extracted:

- Player name
- Position
- Nationality
- Date of birth & age
- Club & league
- Market value
- Injury History

DATA CLEANING

- Removed missing or corrupted rows
- Converted market values from string → numeric
- Standardized country, position, squad, and competition columns
- Extracted and cleaned player ages
- Removed duplicate entries
- Verified data consistency

DATA PRE-PROCESSING

- Created dummy variables for categorical features:
- Nation
- Position
- Squad
- Competition
- Converted monetary fields to **Euro numeric format**
- Normalized features for machine learning models
- Joined multiple scraped datasets into a master dataset

FEATURE ENGINEERING

Important features used:

- Age
- Height
- Position group
- Minutes played
- Goals, assists
- Total appearances
- Previous market trends
- Club strength (competition level)

Created new feature:

- Performance-based ratios
- Form indicators
- Value per minute played

Exploratory Data Analysis

Performed EDA to understand:

- Value distribution
- Relationship between age & market value
- Position-wise value variation
- Influence of player performance metrics
- Outlier players with extremely high values

Time-Series Forecasting with LSTMs

Long Short-Term Memory (LSTM) networks are central to modeling temporal dependencies and forecasting future player values.

1

Univariate LSTM

Basic model on historical performance for future value prediction.

2

Multivariate LSTM

Expanded model with injury history and sentiment for enhanced accuracy.

3

Encoder-Decoder LSTM

Multi-step forecasting for predictions over several transfer windows.

Machine Learning Model

Model used: XGBoost Regressor

Why XGBoost?

- Handles mixed numeric/categorical data
- Robust against missing values
- Excellent performance on tabular data

Model Outcomes:

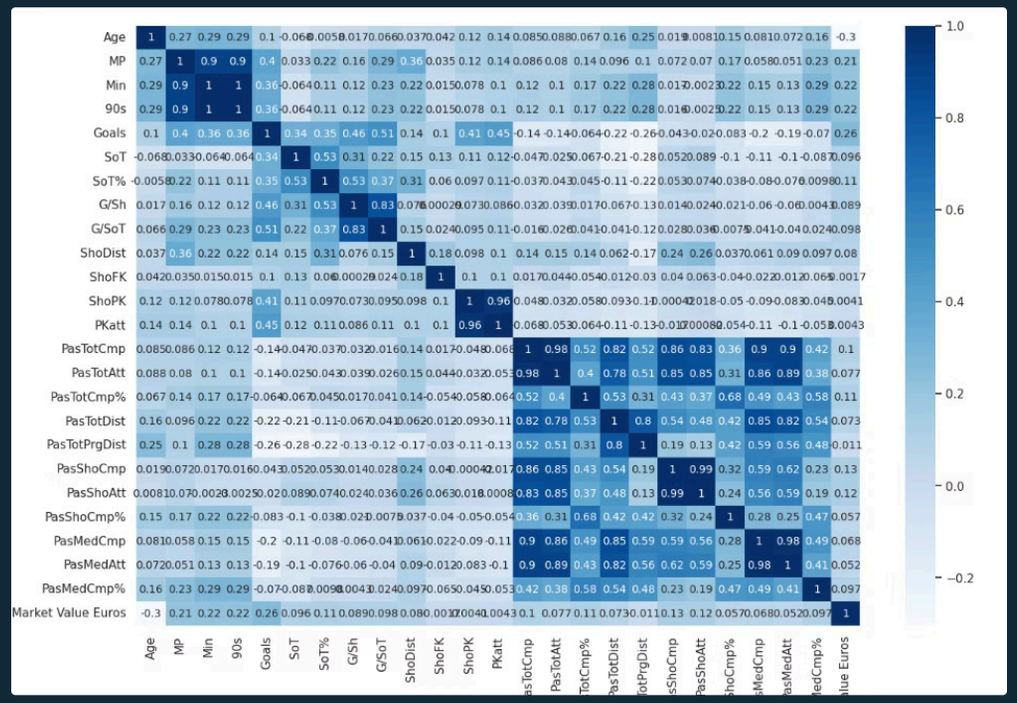
- Trained on cleaned player dataset
- Achieved strong predictive accuracy
- Feature importance identified key determinants of market value

VISUALISATIONS

Market Value by Competition

The bar chart compares **average player market values** across major football leagues.

This visualization highlights how league popularity, spending capacity, and talent demand influence player market valuation.

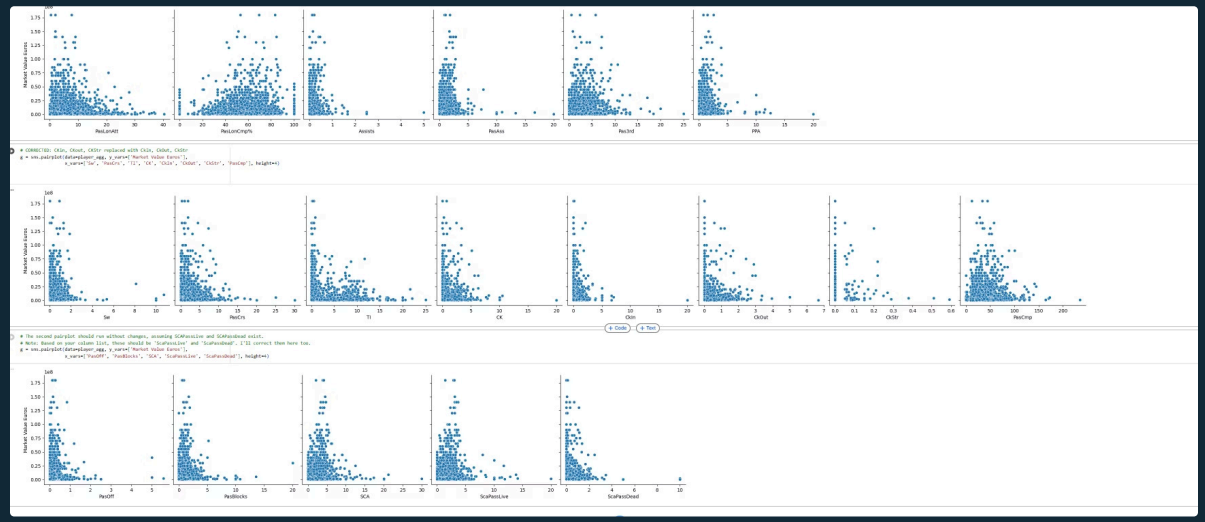


Correlation Heatmap

- The heatmap shows how player performance metrics relate to **Market Value**.
- **Strong correlations** appear with playtime and passing metrics (MP, Min, Total Passes, Progressive Distance).
- Overall, the heatmap highlights which features most strongly drive a player's market value.

Scatterplot Analysis – Performance vs Market Value

- Scatterplots show how different performance stats relate to **Market Value**.
- Most metrics show **weak but positive trends**—higher performance generally increases value.
- Dense clustering at low values indicates most players have lower market values.
- Only a few top performers appear with very high market values.
- Overall, relationships are **scattered and non-linear**.



Model Evaluation

```
Mean Squared Error: 133927566049280.0
R^2 Score on Training: 0.7013140320777893
R^2 Score on Test: 0.3509169816970825
```

METRICS USED:

- **MAE (Mean Absolute Error)**
- **RMSE (Root Mean Squared Error)**
- **R^2 Score**

```
Test Set MSE (Tuned): 137308091187200.0
R^2 Score on Training: 0.9585224390029907
R^2 Score on Test: 0.334533154964447
```

INSIGHTS:

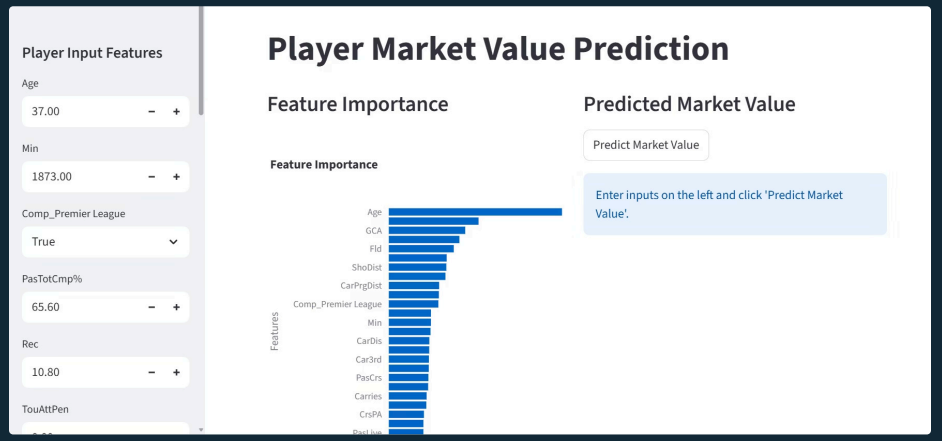
- Age, club competition level, and performance stats heavily influence market value
- Values of top players are harder to predict due to outlier behavior

Deployment (Streamlit App)

Developed a **Streamlit web application**:

- Player info input form
- Real-time value prediction
- Model-based suggestions
- Clean UI for user interaction
- Commands used:

```
pip install streamlit requests beautifulsoup4 pandas xgboost
streamlit run app.py
```



CHALLANGES FACED

- Dynamic webpage content during scraping
- Handling missing player data
- Converting unstructured monetary values
- Model performance on extreme market values

Final Output

1. Automated dataset retrieval
2. End-to-end preprocessing pipeline
3. Machine learning prediction model
4. Interactive web application for predictions
5. Visual insights into player valuation trend